Diabetic Retinopathy

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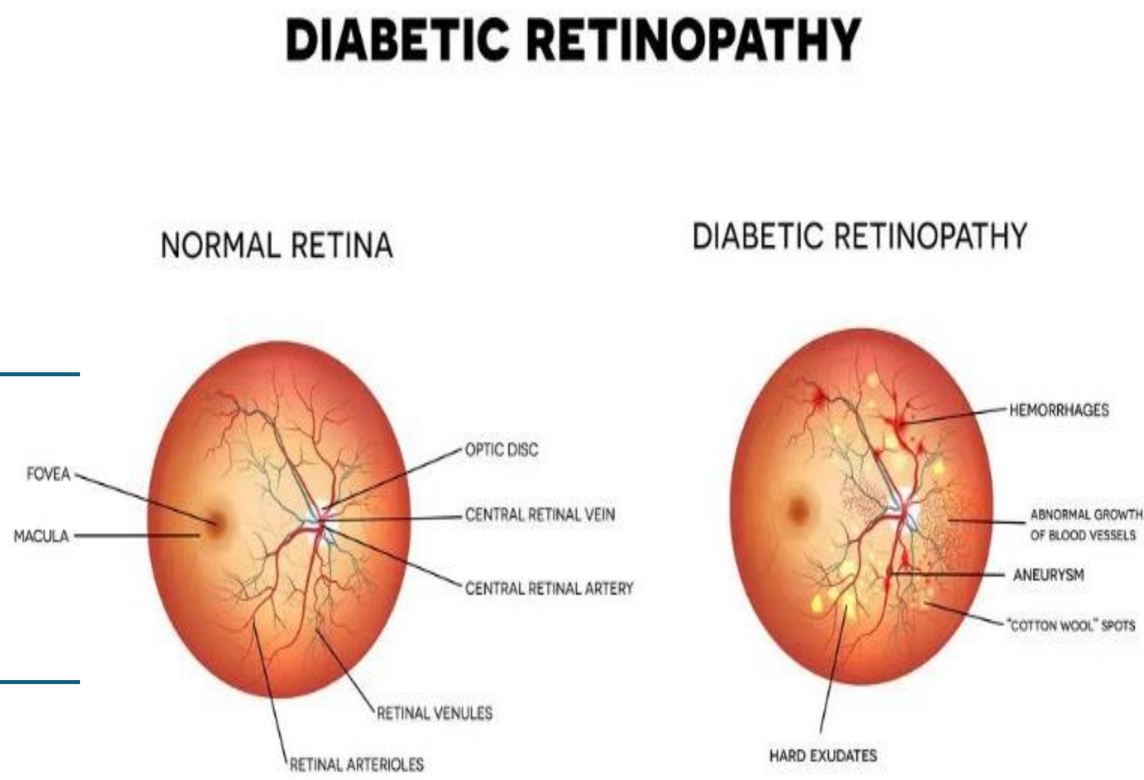
Introduction

What is Diabetic Retinopathy?

Globally, a major cause of visual impairment is diabetic retinopathy, impacting significantly those with chronic diabetes.

Diabetic retinopathy (DR), a complication of diabetes, results from high blood sugar damaging the retina's blood vessels, the light-sensitive tissue at the back of the eye.

Type 1 and **Type 2** diabetics are both at risk of developing diabetic retinopathy.



DR progresses in stages:

1. **Mild Non-Proliferative** – small aneurysms
2. **Moderate** – blockage of some vessels
3. **Severe** – more vessels are blocked, depriving areas of retina
4. **Proliferative** – new, fragile vessels form and may bleed into the eye

Tackling Diabetic Retinopathy: The Problem & Our Purpose

Objective of the project

This project aim uses image processing to automate diabetic retinopathy detection in retinal images. It identifies features like microaneurysms, hemorrhages, and exudates. The project also classifies images by patient diagnostic (DR vs. no DR).

Problem Statement:

- Symptoms of DR are absent until vision damage occurs.
- Manual diagnosis from fundus images is slow, subjective, and expert-dependent.
- Lack of eye care professionals delays diagnoses in resource-poor settings.

Motivation:

Speed up and improve the accuracy of retinal abnormality detection.

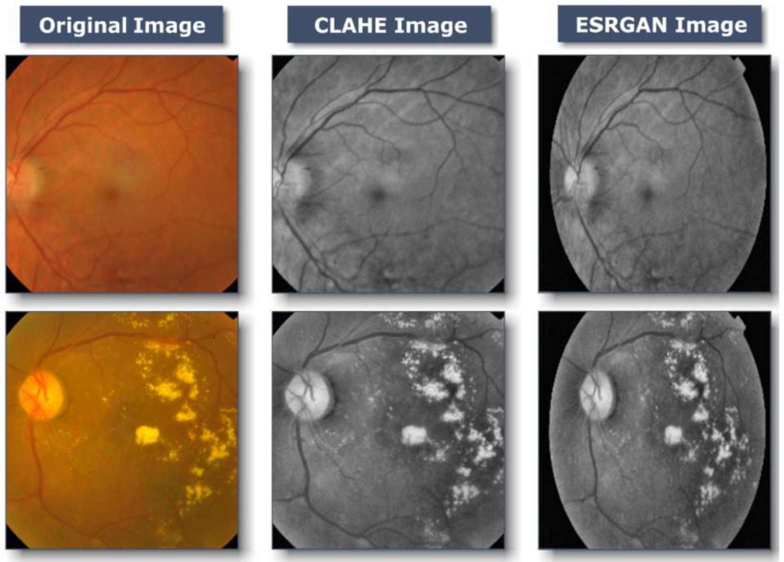
Diminish the workload of healthcare providers.

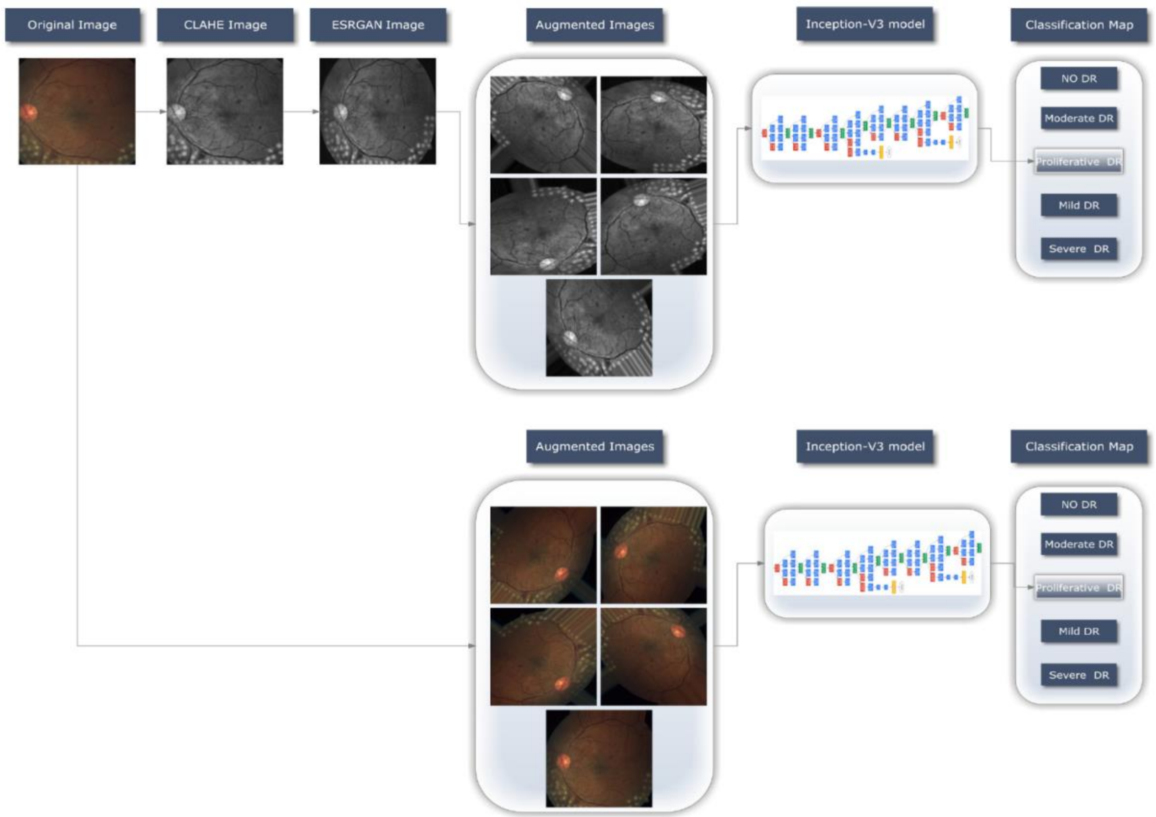
A robust and scalable solution will be created through the application of machine learning (SVM) and (CNN) methodologies.

Early intervention, achieved through expeditious detection, is crucial in preventing blindness.

Deep Learning-Based Prediction of Diabetic Retinopathy Using CLAHE and ESRGAN for Enhancement

- This study utilizes the APTOS 2019 Blindness Detection Dataset, which comprises 3662 high-resolution retinal images, each labeled with one of five DR grades: no DR, mild, moderate, severe, or proliferative DR.
- Classification was performed using the well-regarded Inception-V3 deep learning model. Enhanced preprocessing (Case 1) improved accuracy to 98.7%, exceeding the 80.87% of Case 2.
- CLAHE used tiled histogram equalization for contrast; ESRGAN used RRDBs for detail and noise reduction. The dataset was split to training, validation, and test the enhancements. Adam optimizer tuned hyperparameters across epochs.





Project Approach: Designing the AI Diagnostic Tool

PROGRAMMING LANGUAGE & TOOLS :

- **Python**
- **Tkinter** – GUI Framework
- **NumPy, Matplotlib** – For image manipulation and plotting
- **TensorFlow/Keras** – For DR classification model
- **NumPy** – image arrays and matrix operations
- **Matplotlib** – plot
- **PIL** – safely for image I/O tasks

Classification: (Output = 0: No DR), and (Output = 1: DR)

Design an application with a user-friendly interface that handles automatic in retinal fundus images loading, processing, and result saving.

The app has visually highlight:

- Blood vessels
- Microaneurysms (early DR signs)
- Exudates (indicators of severity)
- FFT and wavelet patterns for texture analysis

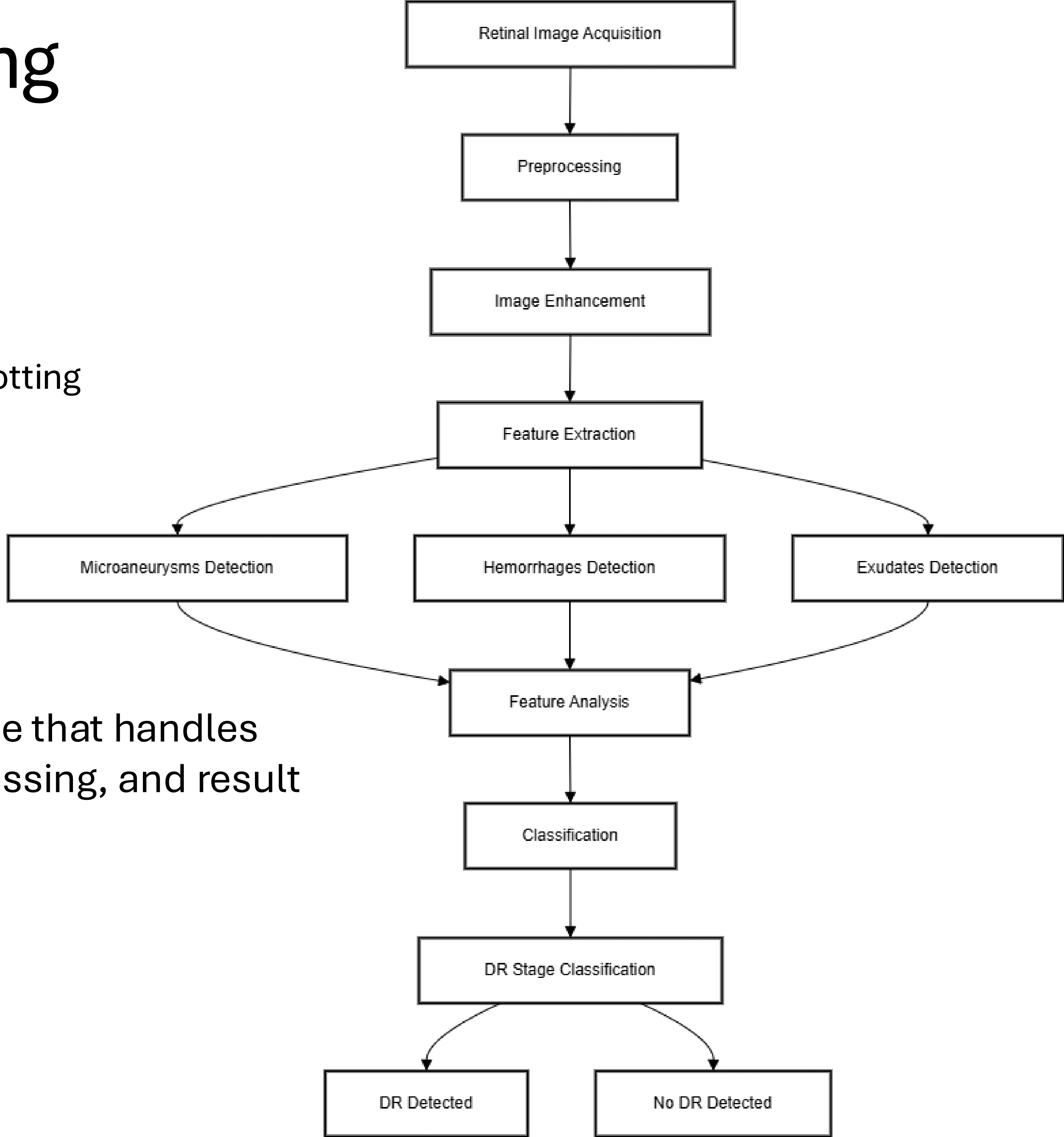


Image Processing Techniques

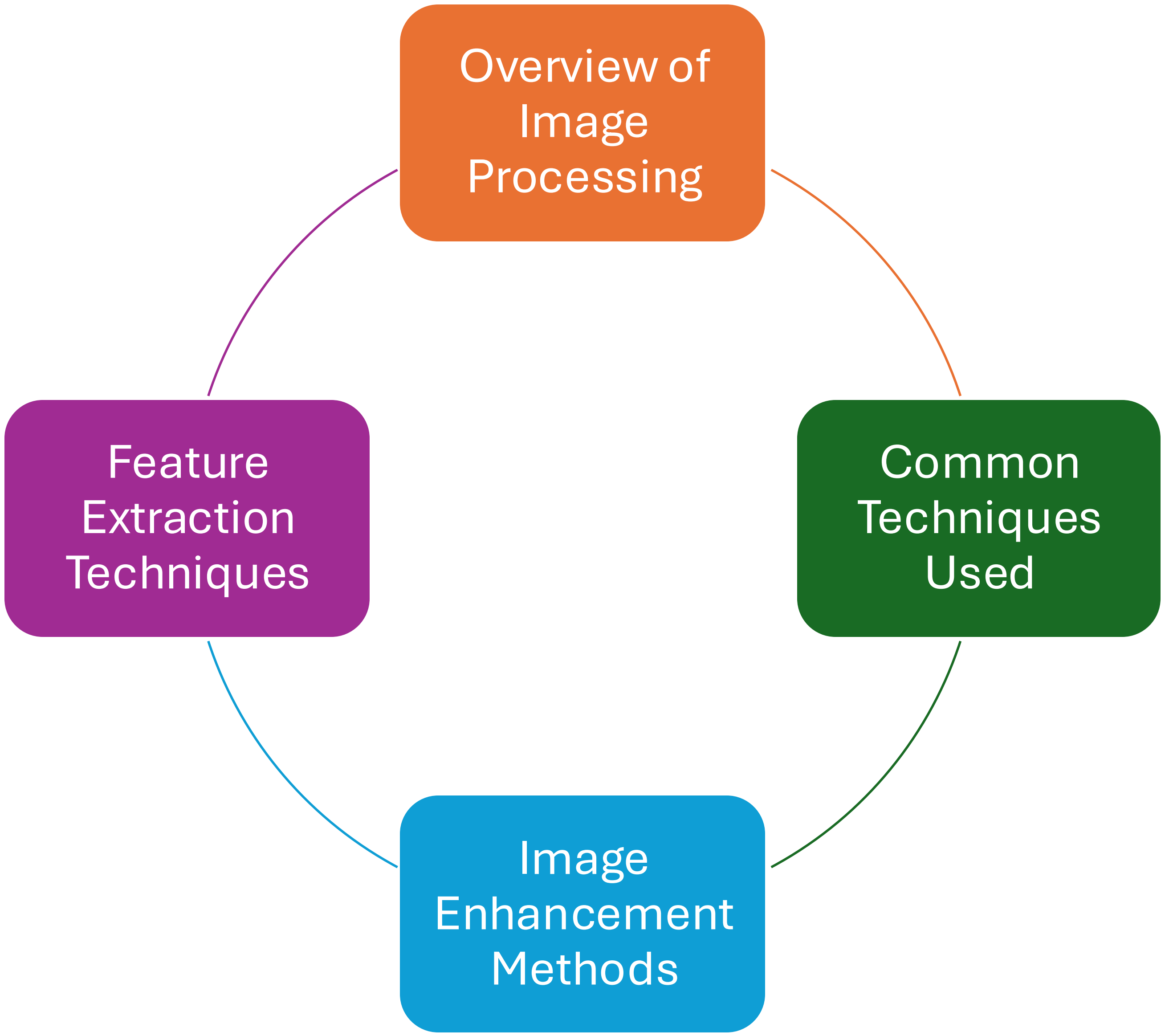


Image Processing

Early diagnosis and monitoring of diabetic retinopathy, a diabetes-related eye complication, significantly benefit from image processing. The automatic detection of retinal image abnormalities improves screening precision and lightens the workload on clinicians.

Noise Reduction & Preprocessing:

- **Gaussian/ Median Filtering**: Gaussian and median filters used to reduce noise and preserve features.

Frequency Analysis:

- **FFT (Fast Fourier Transform):** Analyzes frequency content; useful for enhancing structures like blood vessels or lesions with high-frequency information.

Segmentation Techniques:

- **Thresholding**: Separates objects (e.g., lesions) from background based on intensity.
- **Region Growing (BFS-based)**: Starts from seed points and expands to similar neighbors—good for segmenting connected lesions or vessels.
- **Adaptive Thresholding:** Enhances local contrast and helps identify intensity variations in uneven lighting.
- **Watershed Segmentation:** Uses markers to segment the image; boundaries are detected where different regions meet during flooding.

Edge & Feature Detection:

- **Canny Edge Detection**: Helps guide marker placement for watershed.
- **Morphological Gradient:** Object edges are emphasized by contrasting the effects of dilation and erosion.
- **Skeletonization:** Reduces vessels to a single-pixel width for analysis.

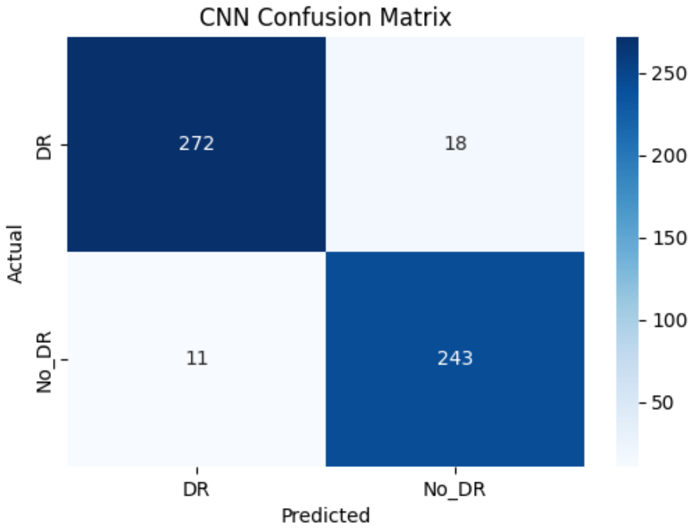
Morphological:

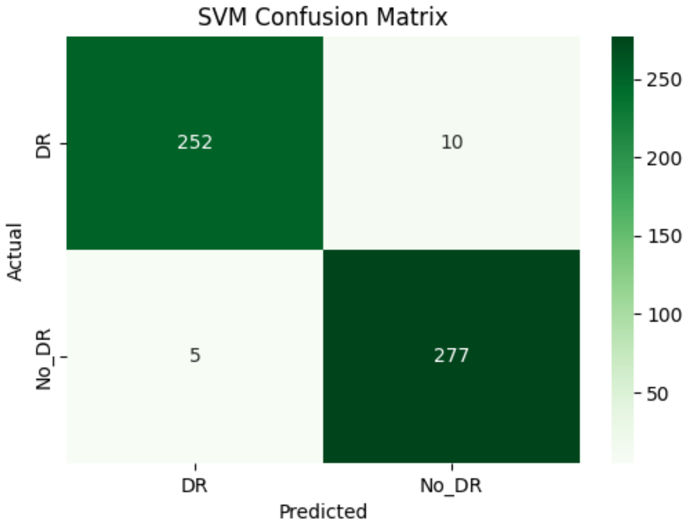
- **Opening: (Erosion + Dilation)**: Removes small noise or detach small objects from larger ones.
- **Closing: (Dilation + Erosion)**: Fills small holes and connects broken parts.
- **Tophat filtering:** Highlights **bright features** smaller than the structuring element.
- **Blackhat filtering:** Highlights **dark features** on a bright background.

Contrast Enhancement :

- **CLAHE:** Boosts contrast locally without over-amplifying noise—ideal for enhancing lesions or vessels.
- **Histogram Equalization:** Globally stretches contrast, improving overall brightness distribution.

Results

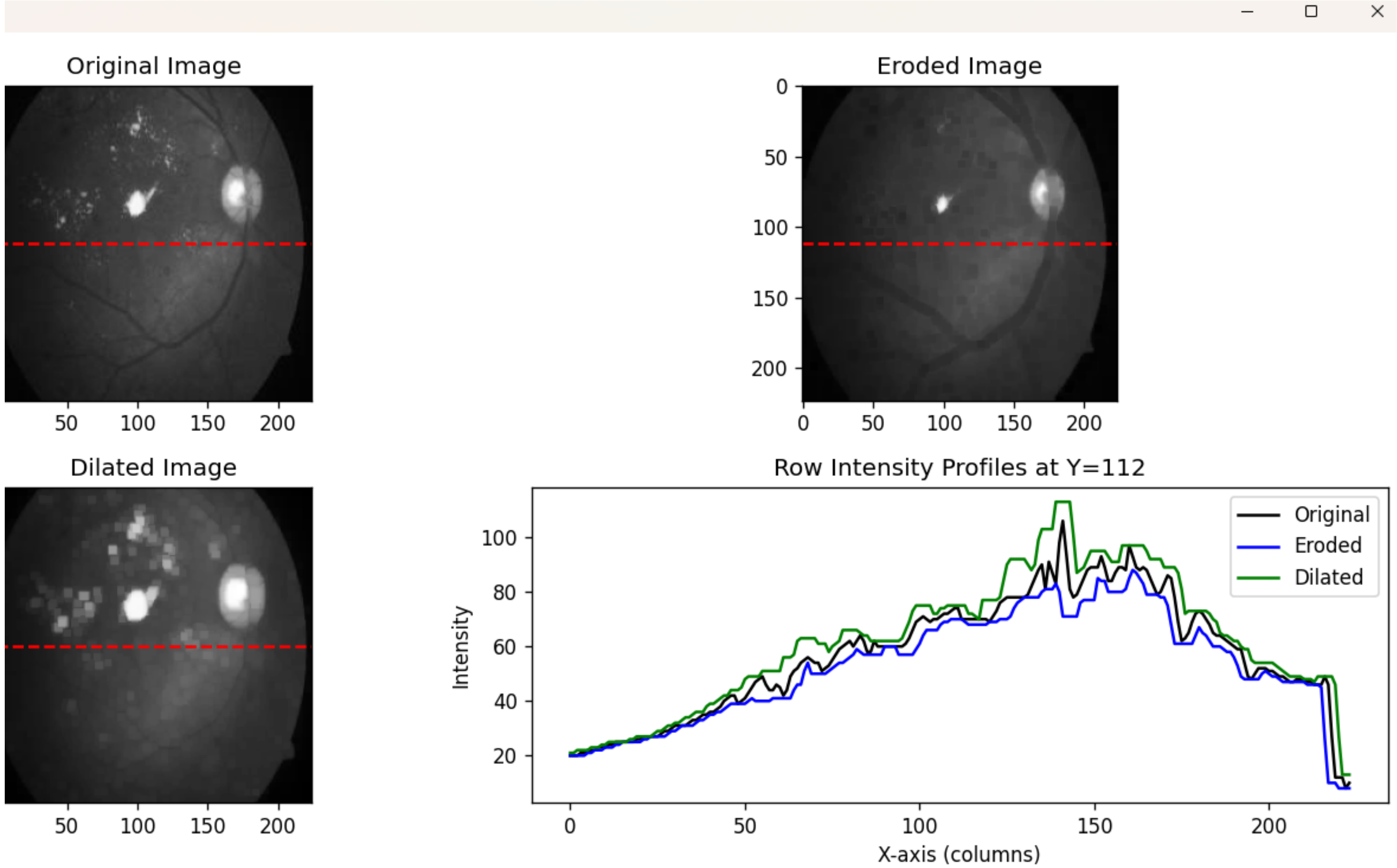




	precision	recall	f1-score	support
DR	0.96	0.94	0.95	290
No_DR	0.93	0.96	0.94	254
accuracy			0.95	544
macro avg	0.95	0.95	0.95	544
weighted avg	0.95	0.95	0.95	544

SVM Classification Report:

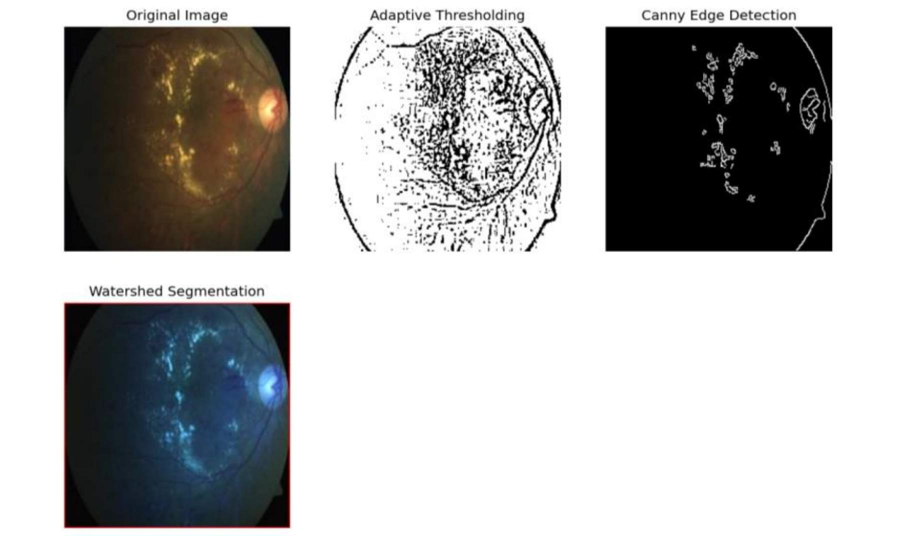
	precision	recall	f1-score	support
DR	0.98	0.96	0.97	262
No_DR	0.97	0.98	0.97	282
accuracy			0.97	544
macro avg	0.97	0.97	0.97	544
weighted avg	0.97	0.97	0.97	544

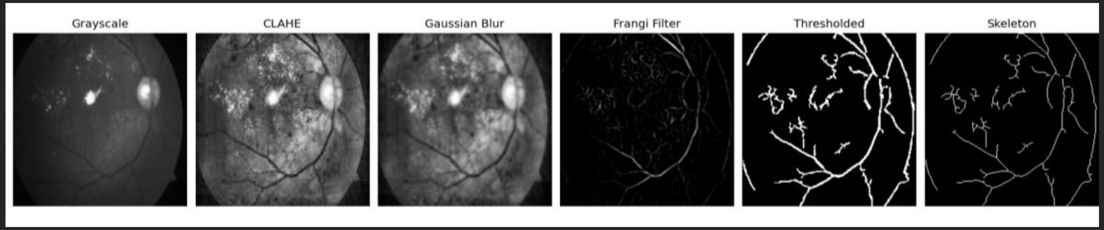


An intensity profile was generated from row Y=112 (indicated by a red dashed line) of the original retinal image (top-left). The eroded image is shown top-right.

Erosion dims bright areas, reducing the visibility of small features like exudates and microaneurysms. Bottom-left: Dilated version.

Bright lesions appear larger and clearer after dilation. A comparison of intensity values along row Y=112 for all three images is shown in the bottom-right plot.


Original Image
Adaptive Thresholding
Canny Edge Detection
Watershed Segmentation


Grayscale
CLAHE
Gaussian Blur
Frangi Filter
Thresholded
Skeleton

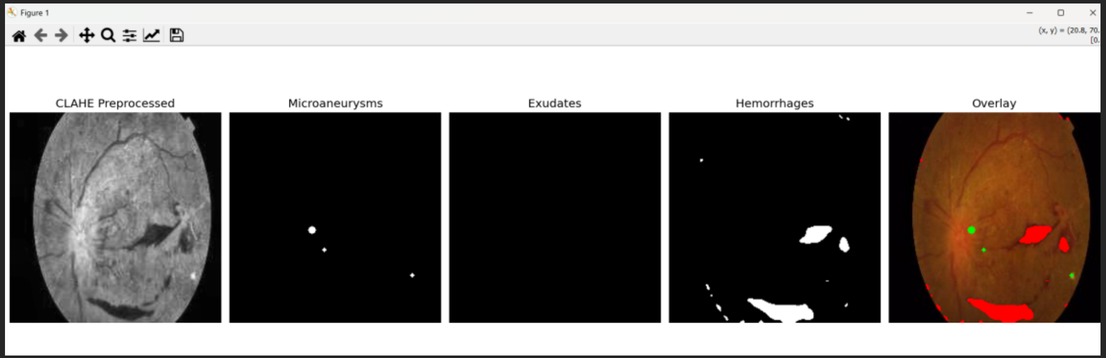
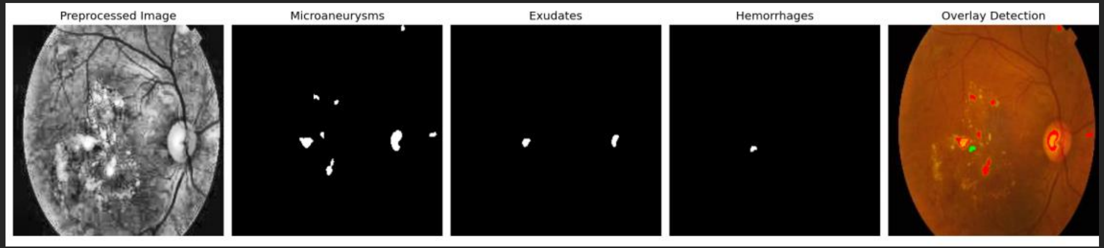
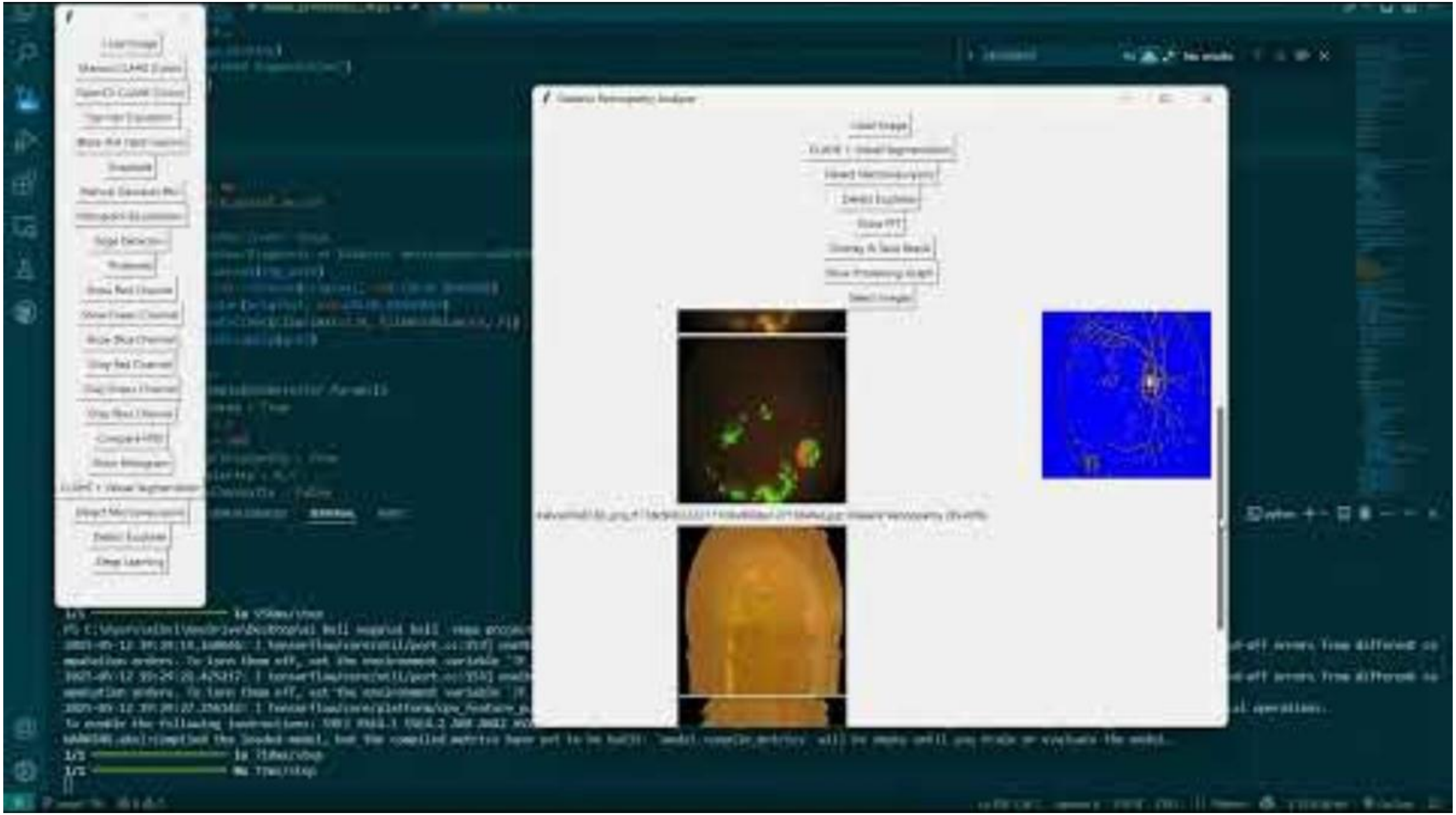


Figure 1
CLAHE Preprocessed
Microaneurysms
Exudates
Hemorrhages
Overlay


Preprocessed Image
Microaneurysms
Exudates
Hemorrhages
Overlay Detection

Link:https://youtu.be/E4Mj-ba0rWc



Q&A

Reference

[1]Alwakid, G., Gouda, W., & Humayun, M. (2023). Deep Learning-Based Prediction of Diabetic Retinopathy Using CLAHE and ESRGAN for Enhancement. *Healthcare*, *11*(6), 863. https://doi.org/10.3390/healthcare11060863

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[3]S. Pavithra, Deepika Jaladi, K. Tamilarasi,

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